Placing Defects Where You Ask: Generating Defects at Chosen Sites in a Model Glass with a Diffusion Model

Midterm report

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Abstract

Defects, small groups of atoms arranged differently from the rest, control how a glass breaks, ages or absorbs light, and studying one needs many realistic copies of it. Today researchers wait for it to form in slow melt-and-cool simulations or insert it by hand, which can leave unnatural surroundings. We train a diffusion model that rebuilds a small region around a chosen site inside an existing glass and is told which defect to create there. On a two-dimensional model glass, over three training runs, it places under-coordinated atoms 3.5 times as often as hand insertion (0.22 vs. 0.06) and B-rich defects 1.5–2.4 times as often (0.45–0.71 vs. 0.30), with atom energies closer to those around natural defects. New since the draft: choosing the guidance strength for each trained model halves the spread between training runs; the rebuilt region leaves a seam at its edge; and placed defects survive heating slightly less often than natural ones.

1 Introduction: motivation and objectives

Glasses have no regular crystal lattice, but they still contain defects, and how a defect behaves depends on the atoms around it. Studies of how glasses deform, age or absorb light therefore need many realistic examples of the same defect at chosen places. In a melt-and-cool (melt-quench) molecular dynamics (MD) simulation one waits for the defect to form by chance, which is slow for rare defects; in hand insertion atoms are moved by hand and relaxed, which is fast but can leave unnatural surroundings. Diffusion models such as DM2 [18] generate realistic glasses but are controlled by whole-sample properties such as the cooling rate, not by a defect at a place.

Like image inpainting [13], the model fills in a region cut out of an existing glass (the rest held fixed) and is told which defect to put at its centre (Task B). Our objectives:

O1 **Placement:** after relaxation, the requested defect is at the site in more of 256 requests than with hand insertion or local melt-quench.

O2 **Realism:** atom energies around a placed defect are close to those around natural defects (local W1 near the natural floor, below hand insertion).

O3 **Reliability:** O1 and O2 hold over three training runs and under changes to the evaluation settings.

We use a two-dimensional model glass, where every step is cheap enough to check carefully, before moving to three-dimensional silica (Section 6).

2 Related work and background

Diffusion models for materials. Diffusion models [9, 15] turn noise into data step by step and now generate crystals [10, 17, 19] and glasses. In **DM2** [18], a network learns to predict random shifts

added to atom positions, and glasses are built by repeating this clean-up with small random kicks. We keep its training target and sampler, use an unlabelled model like DM2's as the "chance" baseline, and follow DM2 in judging realism by comparing structural distributions with MD-made glasses; our local W1 does this around one defect. Other work steers whole samples with spectra or target properties [6, 12].

Local control. RePaint [13] fills a masked image region (our Task A baseline); classifier-free guidance [8] makes a model follow a label more strongly. In materials, the closest work fills in crystals around a fixed host [20] or relaxes crystals with point defects [5]; glasses have no lattice to compare with, so realism must be judged against many natural examples.

3 Data and methods

3.1 Dataset

Source and preparation. We generate our own data with LAMMPS [16], using the two-dimensional Kob–Andersen glass [4, 11]: two atom types, A (large, 65%) and B (small, 35%), interacting through Lennard-Jones forces, in a periodic square box; σ, the size of an A atom, is the unit of length. Preparation: random placement, melt at $T = 2.0$ for 100 time units, cool by 10^{-2} per time unit to $T = 0.01$, minimise the energy, reject glasses that crystallise or contain overlapping atoms or outlier energies, and label defects with rules fixed on the training glasses. The main set has 800/100/100 train/validation/test glasses of $N = 256$ atoms (box side 14.6σ); 1600/200/200 glasses with $N = 64$ serve a symmetry test. An atom's neighbours are the atoms within 1.4 times the contact distance. $\mathrm{D1}^-$ is an A atom with at most 5 neighbours (1.7 per glass), $\mathrm{D1}^+$ an A atom with at least 8 (1.4 per glass), and D2 a B atom with at least two B neighbours within 1.2σ (10 per glass). A library of 10,545 natural defects with their neighbourhoods, cut from the training glasses, supplies hand insertion.

3.2 Model description

We describe the layers of the denoising network (input layer, graph layer, EGNN layers and output layer), then the guidance used when sampling, the architectures we compared (Table 1) and how they were trained.

Input layer. The input represents one glass at noise level σ and must be encoded as numbers. Each of the N atoms has a position $x_i \in \mathbb{R}^2$ in a periodic square box of side L ($N = 256$, $L = 14.6\sigma$), a type $t_i \in \{\mathrm{A}, \mathrm{B}\}$, a flag $m_i \in \{0, 1\}$ that is 1 if the atom is held fixed, and a label token c_i: "unlabelled" everywhere except at the requested site, which carries the requested defect type (or "null" when the label is dropped). The noise level is encoded as 64 random Fourier features of $\log\sigma$ (sines and cosines at fixed random frequencies) followed by a small network. The starting feature vector of atom i is

$$h_i^0 = \mathrm{MLP}\big(E_t(t_i) + E_m(m_i) + E_c(c_i) + \mathrm{MLP}(\mathrm{Fourier}(\log\sigma))\big) \in \mathbb{R}^{128}, \tag{1}$$

where E_t, E_m and E_c are learned look-up tables (embeddings). Positions enter only through the graph, so h_i^0 does not depend on where the glass sits or how it is turned.

Graph layer. Atoms closer than $r_c = 2.5\sigma$ are linked (about 24 neighbours each). The vector r_{ij} from atom i to atom j is the shortest one across the periodic box (minimum image), so atoms near opposite edges can be neighbours. Each distance $|r_{ij}|$ is expanded in 16 Gaussian radial basis functions spread evenly over $[0, r_c]$ and multiplied by a cut-off $f_c(r) = \frac{1}{2}[\cos(\pi r/r_c) + 1]$ that reaches zero at r_c, so links switch on and off smoothly as atoms move.

EGNN layers. Four layers (Figure 1b) pass messages between linked atoms, update the features and produce one vector per atom:

$$\begin{aligned} m_{ij} &= \phi_e\big(h_i, h_j, \mathrm{rbf}(|r_{ij}|)\big)\, f_c(|r_{ij}|), \qquad h_i \leftarrow h_i + \phi_h\Big(\mathrm{LN}(h_i),\ \tfrac{1}{K_0}\textstyle\sum_j g(m_{ij})\, m_{ij}\Big), \\ v_i^{\ell} &= \tfrac{1}{K_0}\textstyle\sum_j r_{ij}\, \phi_x(m_{ij})\, f_c(|r_{ij}|). \end{aligned} \tag{2}$$

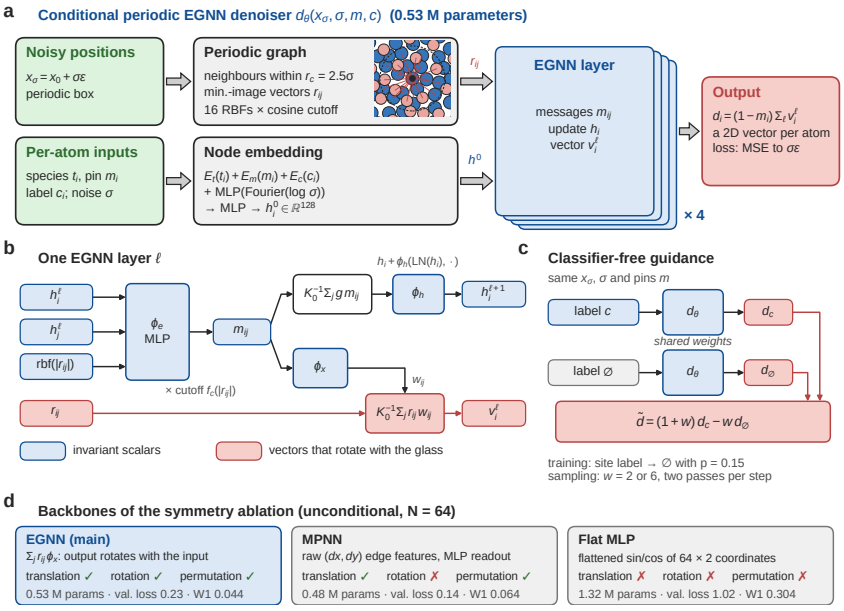


Figure 1: The networks. **(a)** The conditional EGNN denoiser: noisy positions define a periodic graph, per-atom inputs define starting features, and four EGNN layers turn both into one arrow per atom. **(b)** One EGNN layer: blue items do not change when the glass is rotated; red items rotate with it. **(c)** Classifier-free guidance runs the same network with and without the site label. **(d)** The three networks of the symmetry test.

Here ϕ_e ($272 \to 128 \to 128$), ϕ_h ($256 \to 128 \to 128$) and ϕ_x ($128 \to 128 \to 1$) are two-layer networks with SiLU activations, g is a sigmoid gate, LN is layer normalisation and $K_0 = 24$ the typical number of neighbours. Messages depend only on distances, so they do not change when the glass is rotated; v_i^ℓ is a weighted sum of the vectors r_{ij} and turns with the glass. Unlike the original EGNN [14], positions are not moved inside the network. Each layer has 117,890 parameters.

Output layer. The prediction for atom i is $d_i = (1 - m_i)\sum_\ell v_i^\ell \in \mathbb{R}^2$, the network's estimate of how far the noise has moved that atom. It has no activation function (a regression output) and is zero for held atoms. Tests in the repository check numerically that shifting, rotating or renumbering the glass acts on the output as it should.

Classifier-free guidance. When sampling, the same network runs with and without the site label (Figure 1c) and the outputs are combined, $\tilde{d} = (1 + w)\, d_\theta(\cdot, c) - w\, d_\theta(\cdot, \varnothing)$; the weight w sets how strongly the model follows the label [8].

Model architectures. Table 1 summarises the architectures we trained: EGNN_cond, the main model; EGNN_uncond, the same network without labels or held atoms (the chance baseline); and, to test whether built-in symmetry matters, two simpler networks on 64-atom glasses (Figure 1d). The MPNN [7] also receives the raw x and y parts of each neighbour vector, so it handles renumbering but not rotation; the flat MLP reads the sorted coordinates (as sines and cosines), types, flags and labels of all 64 atoms and respects only the periodic box.

Training. The glasses were split into non-overlapping training, validation and test sets in the ratio 0.80:0.10:0.10. The training set was used to fit the weights, the validation set to watch for over-fitting and to choose settings, and the test set only for the final results. The networks were implemented in PyTorch. Following DM2, noise is added to the free atoms, $x_\sigma = x_0 + \sigma\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$ and $\sigma \sim \mathcal{U}(10^{-3}, \sigma_{\max})$, and the network predicts the added shift, minimising the mean squared error over the free atoms U:

$$\mathcal{L}(\theta) = \mathbb{E}\left[\frac{1}{|U|}\sum_{i \in U} \left\| d_\theta(x_\sigma, \sigma, m, c)_i - \sigma\varepsilon_i \right\|^2\right]. \tag{3}$$

Each training example is a whole box (50%), a small patch (30%) or a filled-in region (20%), centred on a natural defect half of the time, and the label is replaced by "null" 15% of the time, which guidance needs. We used the AdamW optimiser with gradient clipping at 1 for 50,000 updates (about

Table 1: Deep learning models; $N \times \{\text{layer}\}$ is a layer repeated N times. Values in brackets were tried. Fixed: 128 features, cutoff 2.5σ, 16 radial basis functions, AdamW, learning rate 2×10^{-4}, batch size 16.

Model	Hidden layers	Parameters / hyper-parameters	Size
EGNN_cond (main)	$1 \times \{$input embedding$\}$, $4 \times \{$EGNN layer$\}$, output sum	training noise $\sigma_{\max}$ (0.5, 1.0); rebuilt region R_{out} (4σ, drawn from 3–6σ); seed (0, 1, 2); guidance w (0, 1, 2, 3, 4, 6, 8); sampling steps (100, 300, 900, 2900)	0.53 M
EGNN_uncond	$1 \times \{$input embedding$\}$, $4 \times \{$EGNN layer$\}$, output sum; no label	training noise $\sigma_{\max}$ (0.5, 1.0); glass size N (64, 256)	0.53 M
MPNN	$1 \times \{$input embedding$\}$, $4 \times \{$MPNN layer$\}$, {Dense 128, Dense 2}	$N = 64$; no label	0.48 M
Flat MLP	$4 \times \{$Dense 512$\}$, Dense 128	$N = 64$; no label	1.32 M

1,000 passes over the 800 training glasses; 40,000 without labels), kept a moving average of the weights (0.999) for sampling, and recorded the validation loss at four fixed noise levels ($\sigma = 0.02$, 0.1, 0.3, 0.5) every 1,000 updates. One run takes about 4 hours on a T4 GPU (Colab or Kaggle).

3.3 Baselines and evaluation

Baselines (implemented in our pipeline). *Chance*: fill the region with no label (DM2-style), showing how often the defect appears anyway. *Hand insertion*, the common practice: paste a natural defect and its neighbours within 1.5σ from a training glass at the site. *Local melt-quench*, the MD way: freeze the atoms beyond R_{out}, then melt and cool the region in LAMMPS. Task A (a whole new box) adds clamping and RePaint [13]. All methods answer the same 256 requests.

Evaluation metrics. *Success*: after relaxation with the FIRE algorithm [2] (all atoms free, largest force below 10^{-6}), the atom of the requested type nearest the site lies within 0.5σ of it and carries the requested label. This is the quantity a user cares about. *Local W1*: the Wasserstein-1 distance between the energies of atoms within $r_{\text{loc}} = 3\sigma$ of the site and the same energies around natural defects of that type in the test glasses. Lower means closer to natural; natural training-glass defects against test-glass defects give the *natural floor* (D2 0.013, $D1^-$ 0.016). We report 95% bootstrap confidence intervals over the 256 samples.

4 Experiments and results

Iterations so far. We trained 13 models to completion (Table 1): two without labels, three conditional runs (seeds 0–2) for each noise range, three networks for the symmetry test and two with a range of region sizes. The first conditional model used DM2's noise range ($\sigma_{\max} = 0.5$), but sampling starts at $\sigma = 1$, a level it had never seen; retraining with $\sigma_{\max} = 1.0$ raised D2 success from 0.54 to 0.79 (seed 0, $w = 6$). We cut the sampler from DM2's 2,900 steps to 100: longer runs over-cool the glass at 15 times the cost. The EGNN made more natural 64-atom glasses than the MPNN (local W1 0.044 vs. 0.064) despite a higher validation loss; the flat MLP did not learn.

Main result and baseline evidence. Table 2 gives Task B over 256 requests per run. Every baseline row comes from our own code on the same requests; we also re-ran DM2's published silica model and confirmed its local structure (99.3% of Si atoms with four O neighbours). The model beats both baselines for D2 and $D1^-$ (O1); at $w = 2$ its local W1 is 1.2–1.5 times the floor for $D1^-$ and 0.045–0.056 for D2 (hand insertion 0.087; O2). Local melt-quench almost never creates the defect: re-melting a 60-atom region is waiting on a small scale. No method places $D1^+$: a natural $D1^+$ moved into another glass with its 47 nearest atoms disappears on relaxation 97% of the time, so what holds it lies further away. The ranking of methods held when we moved the defect rules,

Table 2: Task B: success after relaxation and local W1 (lower is closer to natural). Model rows: mean ± standard deviation over three training runs; baselines: estimate with 95% CI. D2 uses $\sigma_{\max} = 1.0$; D1$^-$ and D1$^+$ use $\sigma_{\max} = 0.5$. †2,900-step sampler. ‡Seed 0 only. §New since the draft: w chosen per model on validation glasses (Section 4.1).

	D2 (B-rich)		D1$^-$ (too few neighbours)		D1$^+$
Method	success	local W1	success	local W1	success
Chance (no label)†	0.03	0.062	0.02	0.053	0.00
Local melt-quench	0.05 [0.03, 0.08]	0.065	0.00 [0.00, 0.00]	0.024	0.01
Hand insertion	0.30 [0.24, 0.36]	0.087	0.06 [0.04, 0.09]	0.032	0.02
Model, $w = 2$	0.45 ± 0.11	**0.051 ± 0.006**	**0.22 ± 0.02**	**0.022 ± 0.002**	0.03‡
Model, $w = 6$	0.71 ± 0.08	0.072 ± 0.016	0.21 ± 0.04	0.038 ± 0.004	0.04‡
Model, w per model§	**0.74 ± 0.04**	0.076 ± 0.013	—	—	—
Natural floor	—	0.013	—	0.016	—

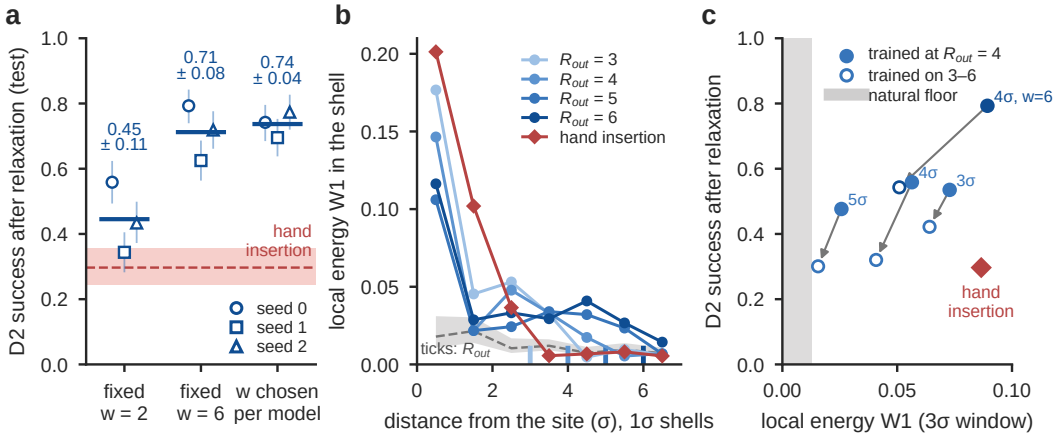


Figure 2: Follow-up checks (Task B, D2). **(a)** Test success for the three training runs at fixed w and at the w each model's validation rows chose (target: validation success ≥ 0.70; chosen $w = 4, 8, 8$). The bar is the mean; the red band is hand insertion. **(b)** Local energy W1 in 1σ shells around the site for regions of radius $R_{\text{out}} = 3$–6σ (seed 0, $w = 2$), hand insertion, and the natural floor (grey); ticks on the axis mark each R_{out}. **(c)** Success against local W1 for the model trained at $R_{\text{out}} = 4\sigma$ (filled) and the model trained on 3–6σ (open), at the same sampling settings (labels give R_{out}; dark: $w = 6$, light: $w = 2$).

success radius, realism window or region size, and re-runs with the same seed reproduced 8 of 9 rows exactly (O3).

4.1 New experiments since the draft

Choosing w for each model. The three D2 runs follow the label with different strength although their validation losses agree within a few percent. We sampled each at $w = 1$–8 on *validation* glasses and applied rules fixed beforehand. The success rule (smallest w with validation success ≥ 0.70) chose $w = 4, 8, 8$; on the test glasses this gives 0.74 ± 0.04, half the spread of fixed $w = 6$ (0.71 ± 0.08), with similar local W1 (Figure 2a). The realism rule (local W1 ≤ 0.07) chose $w = 2$ for every model and changed nothing, because validation W1 runs higher than test W1.

A seam at the edge of the rebuilt region. A region of $R_{\text{out}} = 5\sigma$ had given much lower local W1 than 4σ (0.026 vs. 0.056). Shell by shell (Figure 2b), the mismatch is largest at the defect, stays at about 3 times the floor up to the edge of the rebuilt region, and reaches the floor about 1σ beyond it; hand insertion damages only the first 3σ. A larger region lowers the mismatch near the site, but over a 6σ window every region size scores about the same (W1 0.024–0.026, hand insertion 0.022), so part of the earlier gain was the seam moving out of the 3σ window.

Training on a range of region sizes. Drawing R_{out} from 3–6σ during training left D1$^-$ unchanged

(0.16–0.22 vs. 0.18–0.25). The D2 model, with equal validation loss, followed the label more weakly: fewer successes (0.30–0.54 vs. 0.48–0.79), lower W1 (0.016–0.064 vs. 0.026–0.089; Figure 2c): along the trade-off, not beyond.

D1$^+$ with a larger region. Rebuilding 5–6σ instead of 4σ does not help (success 0.03–0.05): the defect is placed in 24–43% of samples but lost on relaxation.

Physical checks. Table 2 pools failed and successful requests. Successful D2 placements alone ($R_{\text{out}} = 6\sigma$, $w = 2$, seed 0) have local W1 0.029 (natural 0.012, hand insertion 0.094). Host atoms 6–7σ away move by 0.180σ on relaxation (hand insertion 0.064σ). In free MD (56 hosts, three runs each) the defect remains after 250 time units in 87% of generated, 91% of hand-inserted and 95% of natural cases at the dataset's final temperature. Over all 256 requests, 0.48, 0.26 and 0.34 (three runs) end with the defect placed and still present (hand insertion 0.27, no label 0.11).

5 Discussion: challenges and risks

The model meets O1 and O3 for two of three defect types, with lower pooled local W1 than hand insertion (O2). Challenges: runs with equal loss follow the label differently (we now choose w per model); the host is disturbed up to the region's edge and placed defects are a little less stable than natural ones; D1$^+$ needs more than a larger region. The main risk is 3D silica: about 1 s per training update, 565 CPU core-hours of MD, and rare defects (well under one per glass).

6 Next steps

1. **Finish generalisation** (week 10). On fresh glasses of the training kind the runs place D2 in 0.51, 0.36, 0.43 of requests (hand insertion 0.23); 1,024-atom glasses and 10× slower or faster cooling are running.
2. **Start 3D silica** (weeks 9–13): train our conditional NequIP [1] on DM2's 24 glasses (code ready) and make 576-atom glasses with DM2's potential [3], targeting wrongly coordinated Si, non-bridging oxygens and 3-rings.
3. **Fix the weaknesses**: choose w per model by default, anneal the seam shell after sampling, and give D1$^+$ a label for its wider surroundings.

Team contributions

Workstream	What it covers	Members
Data and physics	LAMMPS glass generation, potential and relaxation, local melt-quench baseline	[names]
Defects and evaluation	defect rules, metrics, natural floors, bootstrap statistics, physical checks	[names]
Models and training	EGNN, MPNN and MLP; training runs on Colab and Kaggle	[names]
Sampling and conditioning	sampler, guidance, requests, w selection, seam and region checks	[names]
Baselines and writing	hand insertion, figures, reports, 3D silica plan and compute	[names]

Code and data. Code, datasets, trained models and every result in this report: `https://github.com/chilalaabubakar/IDL-PROJECT`.

References

[1] S. Batzner, A. Musaelian, L. Sun, M. Geiger, J. P. Mailoa, M. Kornbluth, N. Molinari, T. E. Smidt, and B. Kozinsky. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nature Communications*, 13:2453, 2022.

[2] E. Bitzek, P. Koskinen, F. Gähler, M. Moseler, and P. Gumbsch. Structural relaxation made simple. *Physical Review Letters*, 97:170201, 2006.

[3] M. Bouhadja, N. Jakse, and A. Pasturel. Structural and dynamic properties of calcium aluminosilicate melts: a molecular dynamics study. *The Journal of Chemical Physics*, 138:224510, 2013.

[4] R. Brüning, D. A. St-Onge, S. Patterson, and W. Kob. Glass transitions in one-, two-, three-, and four-dimensional binary Lennard-Jones systems. *Journal of Physics: Condensed Matter*, 21(3):035117, 2009.

[5] J. Cui, J. K. Christopher, A. Biswas, P. V. Balachandran, and F. Fioretto. Constrained diffusion for accelerated structure relaxation of inorganic solids with point defects. *arXiv preprint arXiv:2602.19153*, 2026.

[6] J. A. Finkler, Y. Lin, T. Du, J. Hu, and M. M. Smedskjaer. Inverse design of amorphous materials with targeted properties. *Advanced Materials*, 38(39):e22493, 2026.

[7] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl. Neural message passing for quantum chemistry. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1263–1272, 2017.

[8] J. Ho and T. Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

[9] J. Ho, A. Jain, and P. Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.

[10] R. Jiao, W. Huang, P. Lin, J. Han, P. Chen, Y. Lu, and Y. Liu. Crystal structure prediction by joint equivariant diffusion. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[11] W. Kob and H. C. Andersen. Testing mode-coupling theory for a supercooled binary Lennard-Jones mixture I: The van Hove correlation function. *Physical Review E*, 51(5):4626–4641, 1995.

[12] H. Kwon, T. Hsu, W. Sun, W. Jeong, F. Aydin, J. Chapman, X. Chen, V. Lordi, M. R. Carbone, D. Lu, F. Zhou, and T. A. Pham. Spectroscopy-guided discovery of three-dimensional structures of disordered materials with diffusion models. *Machine Learning: Science and Technology*, 5(4):045037, 2024.

[13] A. Lugmayr, M. Danelljan, A. Romero, F. Yu, R. Timofte, and L. Van Gool. RePaint: Inpainting using denoising diffusion probabilistic models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.

[14] V. G. Satorras, E. Hoogeboom, and M. Welling. E(n) equivariant graph neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, pages 9323–9332, 2021.

[15] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

[16] A. P. Thompson et al. LAMMPS – a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Computer Physics Communications*, 271:108171, 2022.

[17] T. Xie, X. Fu, O.-E. Ganea, R. Barzilay, and T. S. Jaakkola. Crystal diffusion variational autoencoder for periodic material generation. In *International Conference on Learning Representations*, 2022.

[18] K. Yang and D. Schwalbe-Koda. A generative diffusion model for amorphous materials. *npj Computational Materials*, 12:29, 2026. doi: 10.1038/s41524-025-01901-1.

[19] C. Zeni, R. Pinsler, D. Zügner, et al. A generative model for inorganic materials design. *Nature*, 639: 624–632, 2025.

[20] P. Zhong, X. Dai, B. Deng, G. Ceder, and K. A. Persson. Crystal structure prediction with host-guided inpainting generation and foundation potentials. *Materials Horizons*, 12(22):9669–9678, 2025.